■ 3.1.7 Low Precision Calculations

When you type in a high precision number, such as `3.00000000000000000000`, *Mathematica* assumes that the number is accurate only to the precision you explicitly give. If you type in a number like `3.`, however, *Mathematica* effectively adds trailing zeroes to give the number some definite "machine precision".

Mathematica effectively adds trailing zeroes, to make a number accurate to the machine precision of 16 digits.

```
In[1]:= Precision[3.]
Out[1]= 16
```

When you work with low precision numbers, *Mathematica* carries out all calculations to the fixed machine precision.

The result is computed to the machine precision, then printed to a default of 6 decimal places.

```
In[2]:= Log[3.]
Out[2]= 1.09861
```

This calculation is again done to the machine precision. When the result is printed, only the first six places are given.

```
In[3]:= Exp[%]
Out[3]= 3.
```

Subtracting 3 gives the difference from the exact result. This difference is again computed to the machine precision.

```
In[4]:= % - 3
                   -16
Out[4]= 4.44089 10
```

The difference has the same precision as your original input.

```
In[5]:= Precision[%]
Out[5]= 16
```

When you do high-precision calculations, *Mathematica* keeps track of the precision you have, and tries to make sure that it gives you results only to the precision that can be justified. This means, for example, that when you subtract two nearby high precision numbers, you typically get a result with much lower precision.

In low precision calculations, *Mathematica* does not explicitly keep track of the precision of your results. Instead, it does *all* computations to a fixed machine precision. This means that *Mathematica* can give you results that go beyond the precision that can be justified in a particular case.

For example, if you subtract two nearby low precision numbers, *Mathematica* gives you a result with the same machine precision. The first few digits of this result will be meaningful. Subsequent digits go beyond the precision that can be justified, and are usually quite meaningless. The digits are included only because *Mathematica* is set up to give a result with a fixed precision.

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The computation is done to the standard machine precision, but the result is printed only to six significant figures.

```
In[6]:= 4. + 10^-12 / 3
Out[6]= 4.
```

The remainder you get on subtracting 4 is not exactly `10^-12/3`. It contains some spurious higher-order digits that go beyond the precision that can be justified.

```
In[7]:= % - 4
                     -13
Out[7]= 3.33067 10
```

The fact that you can get spurious digits in low precision numerical calculations with *Mathematica* is in many respects quite unsatisfactory. The ultimate reason, however, that *Mathematica* uses fixed precision for these calculations is a matter of computational efficiency.

Mathematica is usually set up to insulate you as much as possible from the details of the computer system you are using. In dealing with low precision numbers, you would lose too much, however, if *Mathematica* did not make use of some specific features of your computer.

The important point is that almost all computers have special hardware or microcode for doing floating point calculations to a particular fixed precision. *Mathematica* makes use of these features when doing low precision numerical calculations.

The typical arrangement is that all low precision numbers in *Mathematica* are represented as "double precision floating point numbers" in the underlying computer system. On most current computers, such numbers contain a total of 64 binary bits, typically yielding 16 decimal digits of mantissa.

The main advantage of using the built-in floating point capabilities of your computer is speed. High precision numerical calculations, which do not make such direct use of these capabilities, are usually many times slower than low precision calculations.

There are several disadvantages of using built-in floating point capabilities. One already mentioned is that it forces all numbers to have a fixed precision, independent of what precision can be justified for them.

A second disadvantage is that the treatment of low precision numbers can vary slightly from one computer system to another. In working with low precision numbers, *Mathematica* is at the mercy of the floating point arithmetic system of each particular computer. If floating point arithmetic is done differently on two computers, you may get slightly different results for low precision *Mathematica* calculations on those computers.

One detail to be aware of is that the floating point numbers implemented on typical computers are restricted not only in their precision, but also in their magnitude. If you give *Mathematica* a number whose magnitude lies outside the range

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covered by floating point numbers, *Mathematica* will automatically use its own high precision mechanisms to represent and manipulate the number.

When the result of a computation like this cannot be represented at machine precision, *Mathematica* automatically produces a high precision number.

```
In[8]:= Exp[1000.]

                        434
Out[8]= 1.970071114017 10
```

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